

Fuzzy Relations, Rules and Inferences

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Fuzzy Relations

To understand the fuzzy relations, it is better to discuss first [crisp relation](#).

Suppose, A and B are two (crisp) sets. Then Cartesian product denoted as $A \times B$ is a collection of order pairs, such that

$$A \times B = \{(a, b) | a \in A \text{ and } b \in B\}$$

Note :

(1) $A \times B \neq B \times A$

(2) $|A \times B| = |A| \times |B|$

(3) $A \times B$ provides a mapping from $a \in A$ to $b \in B$.

The mapping so mentioned is called a [relation](#).

Example 1:

Consider the two crisp sets A and B as given below. $A = \{1, 2, 3, 4\}$
 $B = \{3, 5, 7\}$.

Then, $A \times B = \{(1, 3), (1, 5), (1, 7), (2, 3), (2, 5), (2, 7), (3, 3), (3, 5), (3, 7), (4, 3), (4, 5), (4, 7)\}$

Let us define a relation R as $R = \{(a, b) | b = a + 1, (a, b) \in A \times B\}$

Then, $R = \{(2, 3), (4, 5)\}$ in this case.

We can represent the relation R in a matrix form as follows.

$$R = \begin{matrix} & \begin{matrix} 3 & 5 & 7 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Operations on crisp relations

Suppose, $R(x, y)$ and $S(x, y)$ are the two relations define over two crisp sets $x \in A$ and $y \in B$

Union:

$$R(x, y) \cup S(x, y) = \max(R(x, y), S(x, y));$$

Intersection:

$$R(x, y) \cap S(x, y) = \min(R(x, y), S(x, y));$$

Complement:

$$\overline{R(x, y)} = 1 - R(x, y)$$

Example: Operations on crisp relations

Example:

Suppose, $R(x, y)$ and $S(x, y)$ are the two relations defined over two crisp sets $x \in A$ and $y \in B$

$$R = \begin{matrix} & \begin{matrix} 00 & 10 \end{matrix} \\ \begin{matrix} 00 \\ 10 \end{matrix} & \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \end{matrix} \quad \text{and} \quad S = \begin{matrix} & \begin{matrix} 00 & 10 \end{matrix} \\ \begin{matrix} 00 \\ 10 \end{matrix} & \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{matrix};$$

Find the following:

- 1 $R \cup S$
- 2 $R \cap S$
- 3 $\bar{R}$

Composition of two crisp relations

Given R is a relation on X, Y and S is another relation on Y, Z . Then $R \circ S$ is called a composition of relation on X and Z which is defined as follows.

$$R \circ S = \{(x, z) | (x, y) \in R \text{ and } (y, z) \in S \text{ and } \forall y \in Y\}$$

Max-Min Composition

Given the two relation matrices R and S , the **max-min composition** is defined as $T = R \circ S$;

$$T(x, z) = \max\{\min\{R(x, y), S(y, z)\} \text{ and } \forall y \in Y\}$$

Composition: Composition

Example:

Given

$$X = \{1, 3, 5\}; Y = \{1, 3, 5\}; R = \{(x, y) | y = x + 2\}; S = \{(x, y) | x < y\}$$

Here, R and S is on $X \times Y$.

Thus, we have

$$R = \{(1, 3), (3, 5)\}$$
$$S = \{(1, 3), (1, 5), (3, 5)\}$$
$$R = \begin{pmatrix} 3 & 0 & 0 & 1 \\ 5 & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad S = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 3 & 0 & 0 & 1 \\ 5 & 0 & 0 & 0 \end{pmatrix}$$

Using max-min composition $R \circ S =$

Fuzzy relations

- Fuzzy relation is a fuzzy set defined on the Cartesian product of crisp set $X_1, X_2, \dots, X_n$
- Here, n-tuples $(x_1, x_2, \dots, x_n)$ may have varying degree of memberships within the relationship.
- The membership values indicate the strength of the relation between the tuples.

Example:

$X = \{ \text{typhoid, viral, cold} \}$ and $Y = \{ \text{running nose, high temp, shivering} \}$

The fuzzy relation R is defined as

	<i>runningnose</i>	<i>hightemperature</i>	<i>shivering</i>
<i>typhoid</i>	0.1	0.9	0.8
<i>viral</i>	0.2	0.9	0.7
<i>cold</i>	0.9	0.4	0.6

Fuzzy Cartesian product

Suppose

A is a fuzzy set on the universe of discourse X with $\mu_A(x) | x \in X$

B is a fuzzy set on the universe of discourse Y with $\mu_B(y) | y \in Y$

Then $R = A \times B \subset X \times Y$; where R has its membership function given by $\mu_R(x, y) = \mu_{A \times B}(x, y) = \min\{\mu_A(x), \mu_B(y)\}$

Example :

$A = \{(a, 0.2), (a, 0.7), (a, 0.4)\}$ and $B = \{(b, 0.5), (b, 0.6)\}$

$$R = A \times B = \begin{matrix} & & b & b \\ & a & 0.2 & 0.2 \\ & & 0.5 & 0.6 \\ & a & 0.4 & 0.4 \end{matrix}$$

Operations on Fuzzy relations

Let R and S be two fuzzy relations on $A \times B$.

Union:

$$\mu_{R \cup S}(a, b) = \max\{\mu_R(a, b), \mu_S(a, b)\}$$

Intersection:

$$\mu_{R \cap S}(a, b) = \min\{\mu_R(a, b), \mu_S(a, b)\}$$

Complement:

$$\neg \mu_R(a, b) = 1 - \mu_R(a, b)$$

Composition

$$T = R \circ S$$
$$\mu_{R \circ S} = \max_{y \in Y} \{\min\{\mu_R(x, y), \mu_S(y, z)\}\}$$

Operations on Fuzzy relations: Examples

Example:

$$X = (x_1, x_2, x_3); Y = (y_1, y_2); Z = (z_1, z_2, z_3);$$

$$R = \begin{matrix} & \begin{matrix} y_1 & y_2 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 0.5 & 0.1 \\ .2 & 0.9 \\ 0.8 & 0.6 \end{bmatrix} \end{matrix}$$

$$S = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} y_1 \\ y_2 \end{matrix} & \begin{bmatrix} .6 & 0.4 & 0.7 \\ 0.5 & 0.8 & 0.9 \end{bmatrix} \end{matrix}$$

$$R \circ S = \begin{matrix} & \begin{matrix} z_1 & z_2 & z_3 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} & \begin{bmatrix} 0.5 & 0.4 & 0.5 \\ .5 & & 0.9 \\ 0.6 & 0.6 & 0.7 \end{bmatrix} \end{matrix}$$

$$\begin{aligned} \mu_{R \circ S}(x_1, y_1) &= \max\{\min(x_1, y_1), \min(y_1, z_1), \min(x_1, y_2), \min(y_2, z_1)\} \\ &= \max\{\min(0.5, 0.6), \min(0.1, 0.5)\} = \max\{0.5, 0.1\} = 0.5 \text{ and so on.} \end{aligned}$$

Fuzzy relation : An example

Consider the following two sets P and D , which represent a set of paddy plants and a set of plant diseases. More precisely

$P = \{P_1, P_2, P_3, P_4\}$ a set of four varieties of paddy plants

$D = \{D_1, D_2, D_3, D_4\}$ of the four various diseases affecting the plants

In addition to these, also consider another set $S = \{S_1, S_2, S_3, S_4\}$ be the common symptoms of the diseases.

Let, R be a relation on $P \times D$, representing which plant is susceptible to which diseases, then R can be stated as

		D_1	D_2	D_3	D_4
$R =$	P_1	0.6	0.6	0.9	0.8
	P_2	0.1	0.2	0.9	0.8
	P_3	0	.9	0.3	0.4
	P_4	0.9	0.8	0.4	0.2

Fuzzy relation : An example

Also, consider T be the another relation on $D \times S$, which is given by

$$S = \begin{matrix} & \begin{matrix} S_1 & S_2 & S_3 & S_4 \end{matrix} \\ \begin{matrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{matrix} & \begin{bmatrix} 0.1 & 0.2 & 0.7 & 0.9 \\ 1.0 & 1.0 & 0.4 & 0.6 \\ 0.0 & 0.0 & 0.5 & 0.9 \\ 0.9 & 1.0 & 0.8 & 0.2 \end{bmatrix} \end{matrix}$$

Obtain the association of plants with the different symptoms of the disease using **max-min composition**.

Hint: Find $R \circ T$, and verify that

$$R \circ S = \begin{matrix} & \begin{matrix} S_1 & S_2 & S_3 & S_4 \end{matrix} \\ \begin{matrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{matrix} & \begin{bmatrix} 0.8 & 0.8 & 0.8 & 0.9 \\ 0.8 & 0.8 & 0.8 & 0.9 \\ 0.8 & 0.8 & 0.8 & 0.9 \\ 0.8 & 0.8 & 0.7 & 0.9 \end{bmatrix} \end{matrix}$$

Fuzzy relation : Another example

Let, $R = x$ is relevant to y

and $S = y$ is relevant to z

be two fuzzy relations defined on $X \times Y$ and $Y \times Z$, respectively, where $X = \{1, 2, 3\}$, $Y = \{\alpha, \beta, \gamma, \delta\}$ and $Z = \{a, b\}$.

Assume that R and S can be expressed with the following relation matrices :

$$R = \begin{matrix} & \begin{matrix} \alpha & \beta & \gamma & \delta \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0.1 & 0.3 & 0.5 & 0.7 \\ 0.4 & 0.2 & 0.8 & 0.9 \\ 0.6 & 0.8 & 0.3 & 0.2 \end{bmatrix} \end{matrix} \text{ and}$$

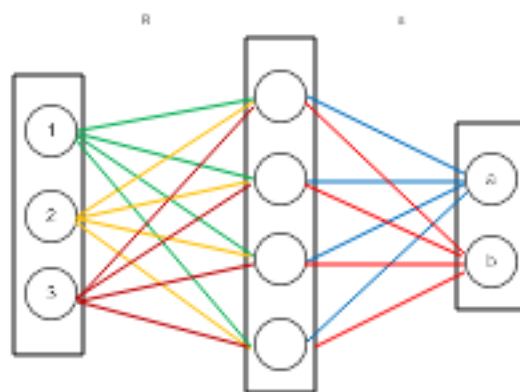
$$S = \begin{matrix} & \begin{matrix} a & b \end{matrix} \\ \begin{matrix} \alpha \\ \beta \\ \gamma \\ \delta \end{matrix} & \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.3 \\ 0.5 & 0.6 \\ 0.7 & 0.2 \end{bmatrix} \end{matrix}$$

Fuzzy relation : Another example

Now, we want to find $R \circ S$, which can be interpreted as a derived fuzzy relation x is **relevant** to z

Suppose, we are only interested in the degree of relevance between $2 \in X$ and $a \in Z$. Then, using max-min composition,

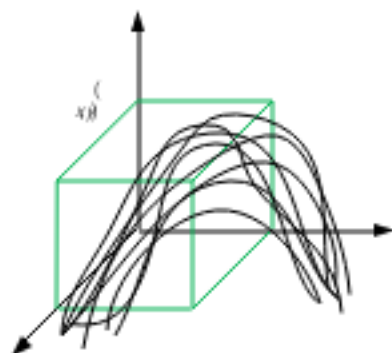
$$\begin{aligned}\mu_{R \circ S}(2, a) &= \max\{(0.4 \wedge 0.9), (0.2 \wedge 0.2), (0.8 \wedge 0.5), (0.9 \wedge 0.7)\} \\ &= \max\{0.4, 0.2, 0.5, 0.7\} = 0.7\end{aligned}$$



2D Membership functions : Binary fuzzy relations

(Binary) fuzzy relations are fuzzy sets $A \times B$ which map each element in $A \times B$ to a membership grade between 0 and 1 (both inclusive).

Note that a membership function of a binary fuzzy relation can be depicted with a 3D plot.



Important: Binary fuzzy relations are fuzzy sets with two dimensional MFs and so on.

2D membership function : An example

Let, $X = R^+ = y$ (the positive real line)
and $R = X \times Y =$ "y is much greater than x"

The membership function of $\mu_R(x, y)$ is defined as

$$\mu_R(x, y) = \begin{cases} \frac{(y-x)}{4} & \text{if } y > x \\ 0 & \text{if } y \leq x \end{cases}$$

Suppose, $X = \{3, 4, 5\}$ and $Y = \{3, 4, 5, 6, 7\}$, then

		3	4		5	6	7
$R =$	3	0	0.25		0.5	0.75	1.0
	4	0	0		0.25	0.5	0.75
	5	0	0		0	0.25	0.5

Problems to ponder:

How you can derive the following? If
x is A or y is B then z is C; Given that

- 1 R_1 : If x is A then z is C [$R_1 \in A \times C$]
- 2 R_2 : If y is B then z is C [$R_2 \in B \times C$]

Hint:

- You have given two relations R_1 and R_2 .
- Then, the required can be derived using the union operation of R_1 and R_2

Fuzzy Propositions

Two-valued logic vs. Multi-valued logic

- The basic assumption upon which crisp logic is based - that every proposition is either TRUE or FALSE.
- The classical two-valued logic can be extended to multi-valued logic.
- As an example, three valued logic to denote true(1), false(0) and indeterminacy ($\frac{1}{2}$).

Two-valued logic vs. Multi-valued logic

Different operations with three-valued logic can be extended as shown in the following truth table:

a	b	$\wedge$	$\vee$	$\neg a$	$\Rightarrow$	$=$
0	0	0	0	1	1	1
0	$\frac{1}{2}$	0	$\frac{1}{2}$	1	1	$\frac{1}{2}$
0	1	0	1	1	1	0
$\frac{1}{2}$	0	0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1
$\frac{1}{2}$	1	$\frac{1}{2}$	1	$\frac{1}{2}$	1	$\frac{1}{2}$
1	0	0	1	1	0	0
1	$\frac{1}{2}$	$\frac{1}{2}$	1	1	$\frac{1}{2}$	$\frac{1}{2}$
1	1	1	1	1	1	1

Fuzzy connectives used in the above table are:

AND ($\wedge$), OR ($\vee$), NOT ($\neg$), IMPLICATION ($\Rightarrow$) and EQUAL ($=$).

Three-valued logic

Fuzzy connectives defined for such a three-valued logic better can be stated as follows:

Symbol	Connective	Usage	Definition
$\neg$	NOT	$\neg P$	$1 - T(P)$
$\vee$	OR	$P \vee Q$	$\max\{T(P), T(Q)\}$
$\wedge$	AND	$P \wedge Q$	$\min\{T(P), T(Q)\}$
$\Rightarrow$	IMPLICATION	$(P \Rightarrow Q)$ or $(\neg P \vee Q)$	$\max\{(1 - T(P)), T(Q)\}$
$=$	EQUALITY	$(P = Q)$ or $[(P \Rightarrow Q) \wedge (Q \Rightarrow P)]$	$1 - T(P) - T(Q) $

Example 1:

P : Ram is honest

- ❶ $T(P) = 0.0$: Absolutely false
- ❷ $T(P) = 0.2$: Partially false
- ❸ $T(P) = 0.4$: May be false or not false
- ❹ $T(P) = 0.6$: May be true or not true
- ❺ $T(P) = 0.8$: Partially true
- ❻ $T(P) = 1.0$: Absolutely true.

Example 2 :Fuzzy proposition

P : Mary is efficient ; $T(P) = 0.8$;

Q : Ram is efficient ; $T(Q) = 0.6$

- ❶ Mary is not efficient.

$$T(\neg P) = 1 - T(P) = 0.2$$

- ❷ Mary is efficient and so is Ram.

$$T(P \wedge Q) = \min\{T(P), T(Q)\} = 0.6$$

- ❸ Either Mary or Ram is efficient

$$T(P \vee Q) = \max\{T(P), T(Q)\} = 0.8$$

- ❹ If Mary is efficient then so is Ram

$$T(P \Rightarrow Q) = \max\{1 - T(P), T(Q)\} = 0.6$$

Fuzzy proposition vs. Crisp proposition

- The fundamental difference between crisp (classical) proposition and fuzzy propositions is in the range of their truth values.
- While each classical proposition is required to be either true or false, the truth or falsity of fuzzy proposition is a matter of degree.
- The degree of truth of each fuzzy proposition is expressed by a value in the interval $[0,1]$ both inclusive.

Canonical representation of Fuzzy proposition

- Suppose, X is a universe of discourse of five persons. Intelligent of $x \in X$ is a fuzzy set as defined below.

Intelligent: $\{(x_1, 0.3), (x_2, 0.4), (x_3, 0.1), (x_4, 0.6), (x_5, 0.9)\}$

- We define a fuzzy proposition as follows:

$P : x$ is intelligent

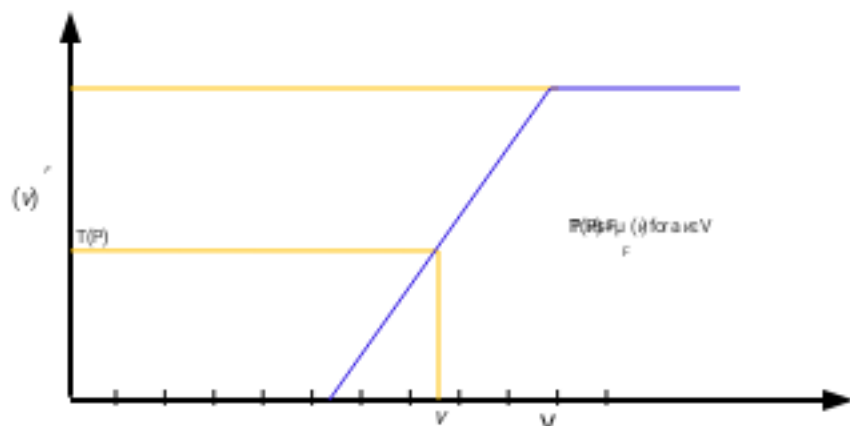
- The canonical form of fuzzy proposition of this type, P is expressed by the sentence $P : v$ is F .

- Predicate in terms of fuzzy set.

$P : v$ is F ; where v is an element that takes values v from some universal set V and F is a fuzzy set on V that represents a fuzzy predicate.

- In other words, given, a particular element v , this element belongs to F with membership grade $\mu_F(v)$.

Graphical interpretation of fuzzy proposition



- For a given value v of variable V in proposition P , $T(P)$ denotes the degree of truth of proposition P .

Fuzzy Implications

- A fuzzy implication (also known as fuzzy If-Then rule, fuzzy rule, or fuzzy conditional statement) assumes the form :

If x is A then y is B

where, A and B are two linguistic variables defined by fuzzy sets A and B on the universe of discourses X and Y , respectively.

- Often, x is A is called the **antecedent** or premise, while y is B is called the **consequence** or conclusion.

Fuzzy implication : Example 1

- If pressure is High then temperature is Low
- If mango is Yellow then mango is Sweet else mango is Sour
- If road is Good then driving is Smooth else traffic is High
- The fuzzy implication is denoted as $R: A \rightarrow B$
- In essence, it represents a binary fuzzy relation R on the (Cartesian) product of $A \times B$

Fuzzy implication : Example 2

- Suppose, P and T are two universes of discourses representing pressure and temperature, respectively as follows.
- $P = \{1, 2, 3, 4\}$ and $T = \{10, 15, 20, 25, 30, 35, 40, 45, 50\}$
- Let the linguistic variable **High temperature** and **Low pressure** are given as
$$\begin{aligned} T_{HIGH} = & \{(20, 0.2), (25, 0.4), (30, 0.6), (35, 0.6), (40, 0.7), (45, 0.8), (50, 0.8)\} \\ P_{LOW} = & \{(1, 0.8), (2, 0.8), (3, 0.6), (4, 0.4)\} \end{aligned}$$

Fuzzy implications : Example 2

- Then the fuzzy implication If temperature is High then pressure is Low can be defined as

$$R: T \rightarrow P$$

HIGH LOW

where, R =

	1	2	3	4
20	0.2		0.2	0.2
25	0.4	0.4	0.4	0.4
30		0.6	0.6	0.6
35	0	0.6		0.4
40	0.7	0.7	0.6	0.4
45	0.8	0.8	0.6	0.4
50	0.8	0.8	0.6	0.4

Note : If temperature is 40 then what about low pressure?

In general, there are two ways to interpret the fuzzy rule $A \rightarrow B$ as A

- coupled with B
- A entails B

Interpretation as **A coupled with B**

$R: A \rightarrow B = A \times B = \int_{X \times Y} \mu_A(x) * \mu_B(y) |_{(x,y)}$; where $*$ is called a **T-norm operator**.

T-norm operator

The most frequently used T-norm operators are:

Minimum: $T_{\min}(a, b) = \min(a, b) = a \wedge b$ **Algebraic**

product: $T_{ap}(a, b) = ab$

Bounded product: $T_{bp}(a, b) = 0 \vee (a + b - 1)$

Drastic product: $T_{dp} = \begin{matrix} a & \text{if } b = 1 \\ 0 & \text{if } a = 1 \\ 0 & \text{if } a, b < 1 \end{matrix}$

Here, $a = \mu_A(x)$ and $b = \mu_B(y)$. T is called the function of T-norm operator.

Based on the T-norm operator as defined above, we can automatically define the fuzzy rule $R: A \rightarrow B$ as a fuzzy set with two-dimensional MF:

$$\mu_R(x, y) = f(\mu_A(x), \mu_B(y)) = f(a, b) \text{ with } a = \mu_A(x), b = \mu_B(y), \text{ and } f \text{ is the fuzzy implication function.}$$

Interpretation as **A coupled with B**

In the following, few implications of $R: A \rightarrow B$

Min operator:

$$R_{\bar{m}} A \times B = \int_{X \times Y} \mu_A(x) \wedge \mu_B(y) \quad \text{or } f(a, b) = a \wedge b$$

[Mamdani rule]

Algebraic product operator

$$R_{ap} A \times B = \int_{X \times Y} \mu_A(x) \cdot \mu_B(y) \quad (xy)^{ap} \text{ or } f(a, b) = ab$$

[Larsen rule]

Bounded product operator

$$\mu_{Bdp} = A \times B = \int_{X \times Y} \mu_A(x) \odot \mu_B(y) |_{(x,y)} = \\ \int_{X \times Y} 0 \vee (\mu_A(x) + \mu_B(y) - 1) |_{(x,y)}$$

$$\text{or } f_{Bdp} = 0 \vee (a + b - 1)$$

Drastic product operator

$$\mu_{Ddp} = A \times B = \int_{X \times Y} \mu_A(x) \wedge \mu_B(y) |_{(x,y)}$$

$$a \text{ if } b = 1$$

$$\text{or } f_{Ddp}(a, b) = \begin{matrix} b & \text{if } a = 1 \\ 0 & \text{if otherwise} \end{matrix}$$

Interpretation of **A entails B**

There are three main ways to interpret such implication:

Material implication :

$$R: A \rightarrow B = A \cup B$$

Propositional calculus :

$$R: A \rightarrow B = A \cup (A \cap B)$$

Extended propositional calculus :

$$R: A \rightarrow B = (A \cap B) \cup B$$

With the above mentioned implications, there are a number of fuzzy implication functions that are popularly followed in fuzzy rule-based system.

Zadeh's arithmetic rule :

$$R_{\bar{z}} A \cup B = \int_{X \times Y} 1 \wedge (1 - \mu^A(x) + \mu^B(y)) | \quad (xy)$$

or

$$f_{za}(a, b) = 1 \wedge (1 - a + b)$$

Zadeh's max-min rule :

$$R_{\bar{m}} A \cup (A \cap B) = \int_{X \times Y} (1 - \mu^A(x)) \vee (\mu^A(x) \wedge \mu^B(y)) |$$

or

$$f_{mm}(a, b) = (1 - a) \vee (a \wedge b)$$

Boolean fuzzy rule

$$R_{BF} A \cup B = \int_{X \times Y} (1 - \mu_A(x)) \vee \mu_B(y) |_{(x,y)}$$

or

$$f_{bf}(a, b) = (1 - a) \vee b$$

Goguen's fuzzy rule:

$$R_{gr} = \int_{X \times Y} \mu_A(x) \star \mu_B(y) |_{xy} \text{ where } a \star b =$$

$$1 \text{ if } a \leq b$$

$$\frac{b}{a} \text{ if } a > b$$

Example 3: Zadeh's Max-Min rule

If x is A then y is B with the implication of Zadeh's max-min rule can be written equivalently as :

$$R_{\text{min}} = (A \times B) \cup (A \times Y)$$

Here, Y is the universe of discourse with membership values for all $y \in Y$ is 1, that is, $\mu_Y(y) = 1 \forall y \in Y$.

Suppose $X = \{a, b, c, d\}$ and $Y = \{1, 2, 3, 4\}$

and $A = \{(a, 0.0), (b, 0.8), (c, 0.6), (d, 1.0)\}$

$B = \{(1, 0.2), (2, 1.0), (3, 0.8), (4, 0.0)\}$ are two fuzzy sets.

We are to determine $R_{\text{min}} = (A \times B) \cup (A \times Y)$

Example 3: Zadeh's min-max rule:

The computation of $R_{nm} = (A \times B) \cup (\bar{A} \times Y)$ is as follows:

$$A \times B =$$

		1	2	3	4	
a		0	0	0		0
	b	0.2	0.8	0.8	0	
c	0.2		0.6	0.6	0	
d	0.2		1.0	0.8	0	

and

$$\bar{A} \times Y =$$

		1	2	3	4	
						1 1 1 1
	b	0.2	0.2	0.2	0.2	
c	0.4		0.4	0.4	0.4	
	d	0	0	0	0	

Example 3: Zadeh's min-max rule:

Therefore,

$$R_{min} = (A \times B) \cup (A \times Y) =$$

		1	2	3	4
a				1	1
	b	0.2	0.8	0.8	0.2
c	0.4		0.6	0.6	0.4
d	0.2		1.0	0.8	0

Example 3 :

$$X = \{a, b, c, d\}$$

$$Y = \{1, 2, 3, 4\}$$

$$\text{Let, } A = \{(a, 0.0), (b, 0.8), (c, 0.6), (d, 1.0)\}$$

$$B = \{(1, 0.2), (2, 1.0), (3, 0.8), (4, 0.0)\}$$

Determine the implication relation :

If x is A then y is B

Here, $A \times B =$

		1	2	3	4	
a						0
		0	0	0		
			b	0.2	0.8	0.8
c	0.2				0.6	0.6
d	0.2				1.0	0.8
						0

Example 3 :

and $A \times Y =$

	1	2	3	4
a	1	1	1	1
b	0.2	0.2	0.2	0.2
c	0.4		0.4	0.4
d	0	0	0	0

$R_{mm} = (A \times B) \cup (A \times Y) =$

	1	2	3	4
a	1	1	1	1
b	0.2	0.8	0.8	0.2
c	0.4	0.6	0.6	0.4
d	0.2	1.0	0.8	0

This R represents If x is A then y is B

Example 3 :

IF x is A THEN y is B ELSE y is C . The relation R is equivalent to

$$R = (A \times B) \cup (\bar{A} \times C)$$

The membership function of R is given by

$$\mu_R(x, y) = \max[\min\{\mu_A(x), \mu_B(y)\}, \min\{\mu_{\bar{A}}(x), \mu_C(y)\}]$$

Example 4:

$$X = \{a, b, c, d\}$$

$$Y = \{1, 2, 3, 4\}$$

$$A = \{(a, 0.0), (b, 0.8), (c, 0.6), (d, 1.0)\}$$

$$B = \{(1, 0.2), (2, 1.0), (3, 0.8), (4, 0.0)\}$$

$$C = \{(1, 0), (2, 0.4), (3, 1.0), (4, 0.8)\}$$

Determine the implication relation :

If x is A then y is B else y is C

Here, $A \times B =$

		1	2	3	4	
a		0	0	0	0	0
	b	0.2	0.8	0.8	0	
c	0.2		0.6	0.6	0	
d	0.2		1.0	0.8	0	

Example 4:

and $A \times C =$

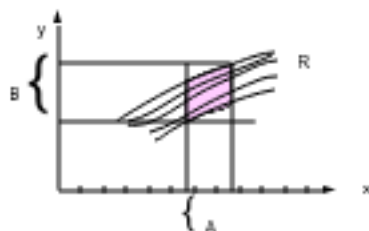
		1	2	3		4
a	0	0	0.4	2	1	0.02 0.8
b	0					0.2
c	0	0.4	0.4	0.4		
d	0	0	0	0		

$R =$

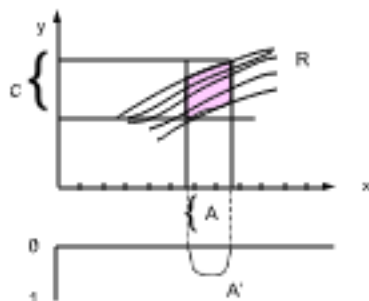
		1	2	3	4
a	0	0.4	1.0	0.8	
	b	0.2	0.8	0.8	0.2
c	0.2		0.6	0.6	0.4
d	0.2		1.0	0.8	0

Interpretation of fuzzy implication

If x is A then y is B



If x is A then y is B else y is C



Fuzzy Inferences

Let's start with propositional logic. We know the following in propositional logic.

- 1 Modus Ponens : $P, P \Rightarrow Q \Leftrightarrow Q$
- 2 Modus Tollens : $P \Rightarrow Q, \neg Q \Leftrightarrow, \neg P$
- 3 Chain rule : $P \Rightarrow Q, Q \Rightarrow R \Leftrightarrow, P \Rightarrow R$

An example from propositional logic

Given

① $C \vee D$

② $\sim H \Rightarrow (A \wedge \sim B)$

③ $C \vee D \Rightarrow \sim H$

④ $(A \wedge \sim B) \Rightarrow (R \vee S)$

From the above can we infer $R \vee S$?

Similar concept is also followed in fuzzy logic to infer a fuzzy rule from a set of given fuzzy rules (also called fuzzy rule base).

Inferring procedures in Fuzzy logic

Two important inferring procedures are used in fuzzy systems :

- Generalized Modus Ponens (GMP)

If x is A Then y is B

x is A

y is B

- Generalized Modus Tollens (GMT)

If x is A Then y is B

y is B

x is A

- Here, A , B , $\hat{A}$ and $\hat{B}$ are fuzzy sets.
- To compute the membership function $\hat{A}$ and $\hat{B}$ the max-min composition of fuzzy sets B and A , respectively with $R(x, y)$ (which is the known implication relation) is to be used.
- Thus,

$$\hat{B} = A \circ R(x, y) \qquad \mu_{\hat{B}}(y) = \max[\min(\mu_A(x), \mu_R(x, y))]$$

$$\hat{A} = B \circ R(x, y) \qquad \mu_{\hat{A}}(x) = \max[\min(\mu_B(y), \mu_R(x, y))]$$

Generalized Modus Ponens

Generalized Modus Ponens (GMP)

P : If x is A then y is B

Let us consider two sets of variables x and y be

$X = \{x_1, x_2, x_3\}$ and $Y = \{y_1, y_2\}$, respectively.

Also, let us consider the following.

$A = \{(x_1, 0.5), (x_2, 1), (x_3, 0.6)\}$

$B = \{(y_1, 1), (y_2, 0.4)\}$

Then, given a fact expressed by the proposition x is A ,

where $A = \{(x_1, 0.6), (x_2, 0.9), (x_3, 0.7)\}$

derive a conclusion in the form y is B (using generalized modus ponens (GMP)).

Example: Generalized Modus Ponens

If x is A Then y is B

x is A

 y is B

We are to find $B' = A' \circ R(x, y)$ where $R(x, y) = \max\{A \times B, A \times Y\}$

$$A \times B = \begin{matrix} & \begin{matrix} y_1 & y_2 & y_3 & y_4 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.5 & 0.4 & 0.5 & 0.5 \\ 1 & 0.4 & 0.4 & 0.4 \end{bmatrix} \end{matrix} \text{ and } A \times Y = \begin{matrix} & \begin{matrix} y_1 & y_2 & y_3 & y_4 \end{matrix} \\ \begin{matrix} x_1 \\ x_2 \end{matrix} & \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0 & 0 & 0 & 0.4 \end{bmatrix} \end{matrix}$$

Note: For $A \times B$, $\mu_{A \times B}(x, y) = \min(\mu_A(x), \mu_B(y))$

Example: Generalized Modus Ponens

$$R(x, y) = (A \times B) \cup (A \times \bar{y}) =$$

	y_1	y_2
x_1	0.5	0.5
x_2	1	0.4
x_3	0.6	0.4

Now, $A = \{(x_1, 0.6), (x_2, 0.9), (x_3, 0.7)\}$

Therefore, $B = A \circ R(x, y) =$

	0.5	0.5
0.6	0.9	0.7
	$\circ 1$	$\circ 0.4$
	0.9	0.5
	0.6	0.4

Thus we derive that y is B where $B = \{(y_1, 0.9), (y_2, 0.5)\}$

Example: Generalized Modus Tollens

Generalized Modus Tollens (GMT)

P: If x is A Then y is B

Q: y is B

 x is A

Example: Generalized Modus Tollens

- Let sets of variables x and y be $X = \{x_1, x_2, x_3\}$ and $Y = \{y_1, y_2\}$, respectively.
- Assume that a proposition **If x is A Then y is B** given where $A = \{(x_1, 0.5), (x_2, 1.0), (x_3, 0.6)\}$ and $B = \{(y_1, 0.6), (y_2, 0.4)\}$
- Assume now that a fact expressed by a proposition **y is B** is given where $B = \{(y_1, 0.9), (y_2, 0.7)\}$.
- From the above, we are to conclude that x is A . That is, we are to determine A .

Example: Generalized Modus Tollens

- We first calculate $R(x, y) = (A \times B) \cup (A \times y)^T$

$$R(x, y) = \begin{matrix} & & \begin{matrix} y & y^c \end{matrix} \\ \begin{matrix} x \\ 0.5 \end{matrix} & \begin{matrix} 0.5 \\ 1 \end{matrix} & \begin{matrix} 0.4 \\ 0.4 \end{matrix} \\ \begin{matrix} x \\ 0.6 \end{matrix} & \begin{matrix} 0.6 \\ 0.4 \end{matrix} & \end{matrix}$$

- Next, we calculate $A' = B \circ R(x, y)$

$$A' = \begin{matrix} & & \begin{matrix} y & y^c \end{matrix} \\ \begin{matrix} x \\ 0.9 \end{matrix} & \begin{matrix} 0.7 \\ 0.5 \end{matrix} & \begin{matrix} 0.5 \\ 0.6 \end{matrix} \\ \begin{matrix} x \\ 0.6 \end{matrix} & \begin{matrix} 0.6 \\ 0.4 \end{matrix} & \end{matrix} \circ \begin{matrix} \begin{matrix} x \\ 0.5 \end{matrix} & \begin{matrix} 0.5 \\ 1 \end{matrix} \\ \begin{matrix} x \\ 0.6 \end{matrix} & \begin{matrix} 0.4 \\ 0.4 \end{matrix} \end{matrix} = \begin{matrix} \begin{matrix} x \\ 0.5 \end{matrix} & \begin{matrix} 0.9 \end{matrix} \\ \begin{matrix} x \\ 0.6 \end{matrix} & \begin{matrix} 0.6 \end{matrix} \end{matrix}$$

- Hence, we calculate that x is A' where $A' = [(x_1, 0.5), (x_2, 0.9), (x_3, 0.6)]$

Apply the fuzzy GMP rule to deduce **Rotation is quite slow**

Given that :

- If temperature is High then rotation is Slow.
- temperature is Very High

Let,

$X = \{30, 40, 50, 60, 70, 80, 90, 100\}$ be the set of temperatures.

$Y = \{10, 20, 30, 40, 50, 60\}$ be the set of rotations per minute.

The fuzzy set High(H), Very High (VH), Slow(S) and Quite Slow (QS) are given below.

$$H = \{(70, 1), (80, 1), (90, 0.3)\}$$

$$VH = \{(90, 0.9), (100, 1)\}$$

$$S = \{(30, 0.8), (40, 1.0), (50, 0.6)\}$$

$$QS = \{(10, 1), (20, 0.8)\}$$

- ❶ If temperature is High then the rotation is Slow.

$$R = (H \times S) \cup (H \times \bar{Y})$$

- ❷ temperature is Very High

Thus, to deduce "rotation is Quite Slow", we make use the composition rule $QS = VH \circ R(x, y)$

8. s' [4 and 7, Modus Tollens]

Problem 3.12. Consider the fuzzy rule R : If *service is good* Then *customer is satisfied*. Related universes are *service-rating* = $\{a, b, c, d, e\}$, and *satisfaction-grade* = $\{1, 2, 3, 4, 5\}$ where the service-ratings a, b, c, d, e are in descending order and the satisfaction-grades 1, 2, 3, 4, 5 are in the ascending order. The fuzzy sets *good-service* and *satisfied* are defined as follows:

$$\text{good-service} = \frac{1.0}{a} + \frac{0.8}{b} + \frac{0.6}{c} + \frac{0.4}{d} + \frac{0.2}{e}$$

$$\text{satisfied} = \frac{0.2}{1} + \frac{0.4}{2} + \frac{0.6}{3} + \frac{0.8}{4} + \frac{1.0}{5}.$$

Find the relation matrix for this rule according to Zadeh's interpretation.

Solution 3.12. According to Zadeh's interpretation, the rule R is expressed by the relation

$$R = (\text{good-service} \times \text{satisfied}) \cup (\overline{\text{good-service}} \times \text{satisfaction-grade})$$

The computation of the relation R representing the information contained in the fuzzy rule stated above is given below.

$$\text{good-service} \times \text{satisfied} = \begin{matrix} & \begin{matrix} 5 & 4 & 3 & 2 & 1 \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 1.0 & 0.8 & 0.6 & 0.4 & 0.2 \\ 0.8 & 0.8 & 0.6 & 0.4 & 0.2 \\ 0.6 & 0.6 & 0.6 & 0.4 & 0.2 \\ 0.4 & 0.4 & 0.4 & 0.4 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix} \end{matrix},$$

$$\overline{\text{good-service} \times \text{satisfaction-grade}} = \begin{matrix} & \begin{matrix} 5 & 4 & 3 & 2 & 1 \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0.4 & 0.4 & 0.4 & 0.4 & 0.4 \\ 0.6 & 0.6 & 0.6 & 0.6 & 0.6 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \end{bmatrix} \end{matrix},$$

Therefore, $R = (\text{good-service} \times \text{satisfied}) \cup (\overline{\text{good-service} \times \text{satisfaction-grade}})$

$$= \begin{matrix} & \begin{matrix} 5 & 4 & 3 & 2 & 1 \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 1.0 & 0.8 & 0.6 & 0.4 & 0.2 \\ 0.8 & 0.8 & 0.6 & 0.4 & 0.2 \\ 0.6 & 0.6 & 0.6 & 0.4 & 0.4 \\ 0.6 & 0.6 & 0.6 & 0.6 & 0.6 \\ 0.8 & 0.8 & 0.8 & 0.8 & 0.8 \end{bmatrix} \end{matrix}$$

Problem 3.13. Recall the rule *If job is risky Then compensation is high* discussed in Example 3.19. The related fuzzy sets are

$$\text{risky-job} = \frac{0.3}{\text{job}_1} + \frac{0.8}{\text{job}_2} + \frac{0.7}{\text{job}_3} + \frac{0.9}{\text{job}_4}$$

$$\text{high-compensation} = \frac{0.2}{c_1} + \frac{0.4}{c_2} + \frac{0.6}{c_3} + \frac{0.8}{c_4}$$

on the universes $\text{job} = \{\text{job}_1, \text{job}_2, \text{job}_3, \text{job}_4\}$ and $\text{compensation} = \{c_1, c_2, c_3, c_4\}$. Now, let the premise be '*job is risky*' where the predicate risky_1 is represented by the fuzzy set

$$\text{risky}_1\text{-job} = \frac{0.3}{\text{job}_1} + \frac{1.0}{\text{job}_2} + \frac{1.0}{\text{job}_3} + \frac{0.2}{\text{job}_4}$$

Employing Generalized Modus Ponens, we reach the conclusion '*compensation is high*'. Compute $\text{high}_1\text{-compensation}$.

Solution 3.13. The rule R : If *job is risky* Then *compensation is high* is represented by the fuzzy relation

$$R = (\text{risky-job} \times \text{high-compensation}) \cup (\overline{\text{risky-job}} \times \text{compensation})$$

Which, on necessary calculations, is expressed by the fuzzy relation

$$R = \begin{matrix} & \begin{matrix} C_1 & C_2 & C_3 & C_4 \end{matrix} \\ \begin{matrix} job_1 \\ job_2 \\ job_3 \\ job_4 \end{matrix} & \begin{bmatrix} 0.7 & 0.7 & 0.7 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.3 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \end{bmatrix} \end{matrix}$$

Now, the fuzzy set $high_1\text{-compensation} = risky_1\text{-job} \circ R$ is obtained as

$$[0.3 \ 1.0 \ 1.0 \ 0.2] \circ \begin{bmatrix} 0.7 & 0.7 & 0.7 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.3 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \end{bmatrix} \\ = [0.3 \ 0.4 \ 0.6 \ 0.8]$$

Hence,

$$high_1\text{-compensation} = \frac{0.3}{c_1} + \frac{0.4}{c_2} + \frac{0.6}{c_3} + \frac{0.8}{c_4}$$

$$u_1 \quad u_2 \quad u_3 \quad u_4$$

Problem 3.14. Let $U = V = \{0, 1, 2, 3, 4\}$ be the universe of discourse on which the fuzzy set $small = \frac{1.0}{0} + \frac{0.5}{1} + \frac{0.2}{2} + \frac{0.1}{3} + \frac{0.0}{4}$ is defined. Again, let R be the fuzzy relation 'more or less the same' which is defined by the relation matrix shown below.

$$R = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1.0 & 0.5 & 0.1 & 0.0 & 0.0 \\ 0.5 & 1.0 & 0.5 & 0.1 & 0.0 \\ 0.1 & 0.5 & 1.0 & 0.5 & 0.1 \\ 0.0 & 0.1 & 0.5 & 1.0 & 0.5 \\ 0.0 & 0.0 & 0.1 & 0.5 & 1.0 \end{bmatrix} \end{matrix}$$

If the premise and the rule are stated as

Premise: x is small.

Rule: x is more or less the same as y

then apply a suitable fuzzy rule of inference to obtain the conclusion and express it suitably as a relation.

Any questions??